

CONVOLUTIONAL NEURAL NETWORK

ROAD SIGNS CLASSIFICATION

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What is Convolutional Neural Network?

- CONVOLUTIONAL NEURAL NETWORK(CNN)IS A TYPE OF DEEP LEARNING ALGORITHM SPECIFICALLY DESIGNED TO ANALYZE AND RECOGNIZE IMAGES.
- IT AUTOMATICALLY DETECTS IMPORTANT FEATURES (EDGES, COLORS, SHAPES, TEXTURES) IN AN IMAGE — SO YOU DON'T NEED TO MANUALLY EXTRACT FEATURES.
- THINK OF CNNs AS GIVING A COMPUTER EYES TO SEE AND UNDERSTAND VISUALS.



DATASET OVERVIEW

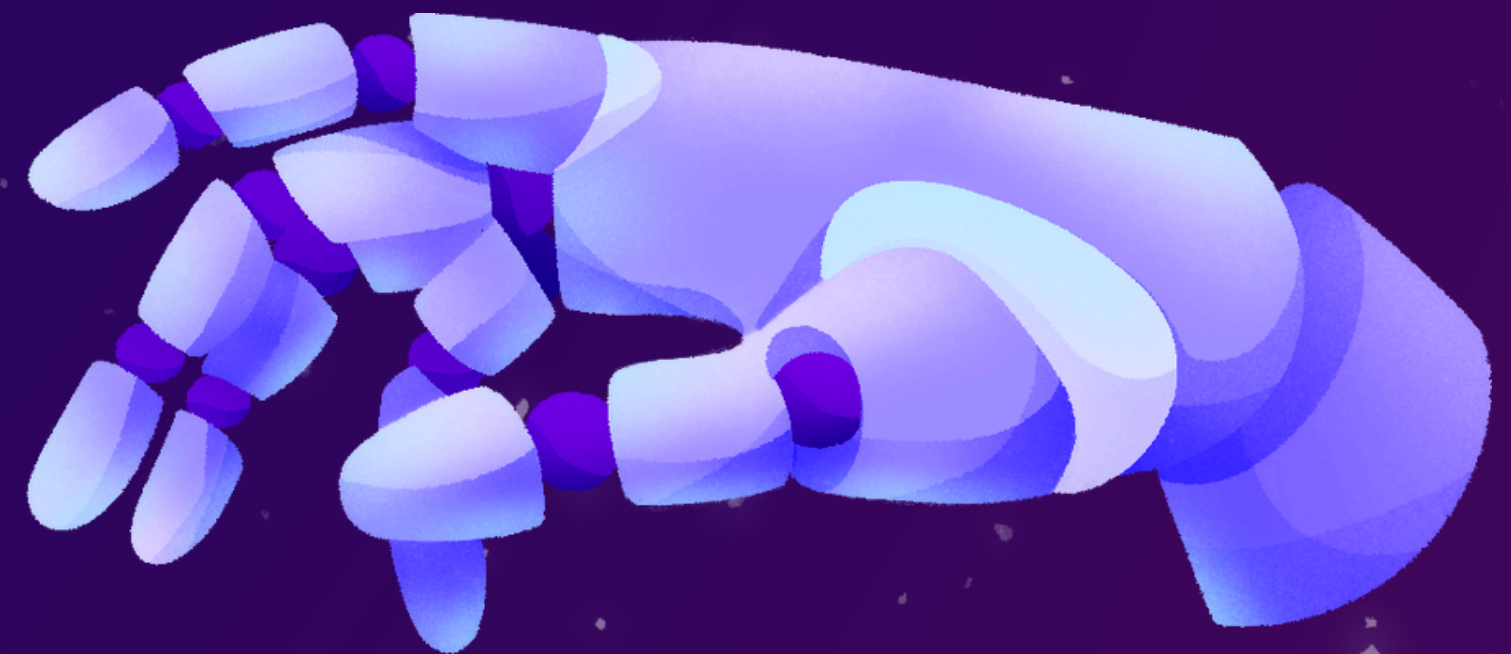
- ROAD SIGN RECOGNITION (RSR) IS A COMPUTER VISION TASK WHERE A SYSTEM AUTOMATICALLY DETECTS AND CLASSIFIES TRAFFIC SIGNS FROM IMAGES OR VIDEOS — JUST LIKE WHAT SELF-DRIVING CARS DO.
- IT HELPS THE VEHICLE (OR SOFTWARE) UNDERSTAND ROAD RULES SUCH AS:

 STOP

 CAUTION

 SPEED LIMIT (30, 50, ETC.)

 TURN / U-TURN



DATA COLLECTION AND PREPROCESSING

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LABELS.CSV LOADED SUCCESSFULLY
COLUMNS-['CLASSID', 'NAME']

LOADED IMAGES: (743, 64, 64, 3)

 **LOADED LABELS: (743,)**

FOUND 743 IMAGES BELONGING TO 31 CLASSES.

FOUND 285 IMAGES BELONGING TO 30 CLASSES.

BUILD THE (CNN)

```
MODEL = SEQUENTIAL([  
    CONV2D(32, (3,3), ACTIVATION='RELU', INPUT_SHAPE=(IMG_SIZE, IMG_SIZE, 3)),  
    MAXPOOLING2D(2,2),  
    CONV2D(64, (3,3), ACTIVATION='RELU'),  
    MAXPOOLING2D(2,2),  
    FLATTEN(),  
    DENSE(128, ACTIVATION='RELU'),  
    DROPOUT(0.5),  
    DENSE(LEN(NP.UNIQUE(LABELS)), ACTIVATION='SOFTMAX')
```

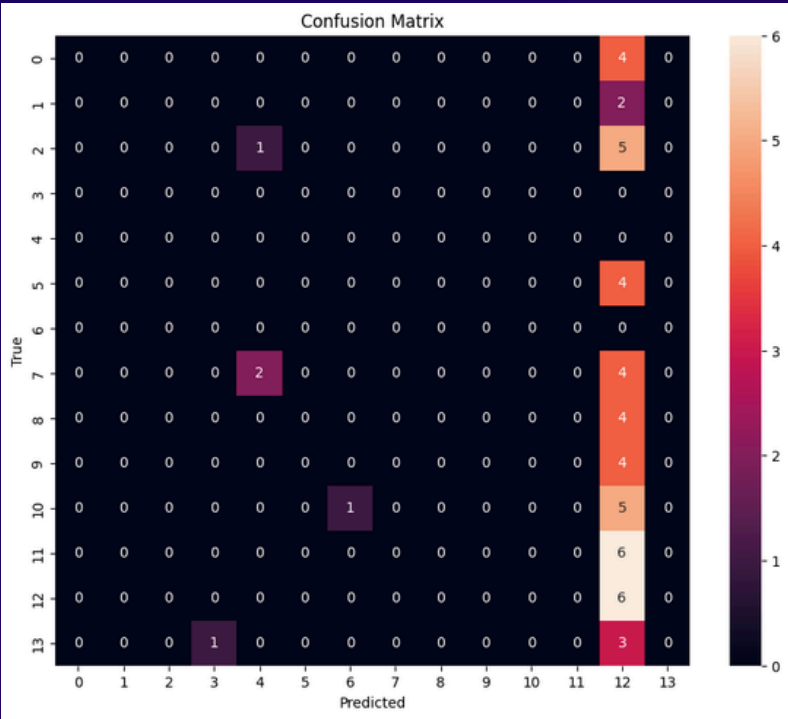
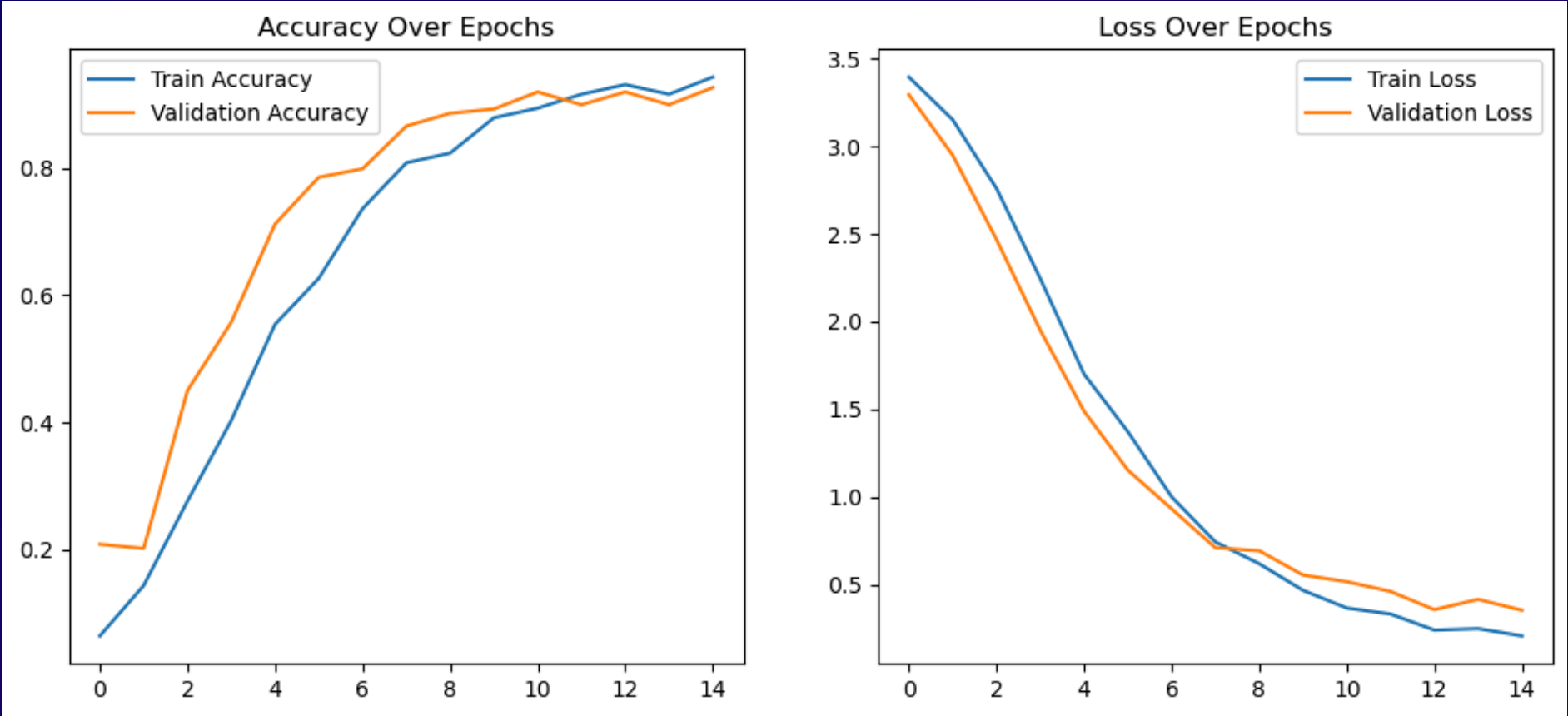
```
MODEL.COMPILE(OPTIMIZER='ADAM', LOSS='CATEGORICAL_CROSSENTROPY', METRICS=['ACCURACY'])
```

COMPILATION OF THE MODEL

LAYER (TYPE)	OUTPUT SHAPE	PARAM #
CONV2D (CONV2D)	(NONE, 62, 62, 32)	896
MAX_POOLING2D (MAXPOOLING2D)	(NONE, 31, 31, 32)	0
CONV2D_1 (CONV2D)	(NONE, 29, 29, 64)	18,496
MAX_POOLING2D_1 (MAXPOOLING2D)	(NONE, 14, 14, 64)	0
FLATTEN (FLATTEN)	(NONE, 12544)	0
DENSE (DENSE)	(NONE, 128)	1,605,...
DROPOUT (DROPOUT)	(NONE, 128)	0
DENSE_1 (DENSE)	(NONE, 30)	3,870

MODEL EVALUATION AND TESTING

Output predictions and compare them with actual labels



	precision	recall	f1-score	support
0	0.00	0.00	0.00	4
1	0.00	0.00	0.00	6
2	0.00	0.00	0.00	12
3	0.00	0.00	0.00	22
4	0.00	0.00	0.00	9
5	0.00	0.00	0.00	11
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	15
8	0.00	0.00	0.00	11
9	0.00	0.00	0.00	19
10	0.00	0.00	0.00	10
11	0.00	0.00	0.00	10
12	0.00	0.00	0.00	10
13	0.00	0.00	0.00	2
14	0.00	0.00	0.00	8
15	0.00	0.00	0.00	5
16	0.04	0.14	0.06	7
17	0.00	0.00	0.00	10
18	0.00	0.00	0.00	2
19	0.10	0.08	0.09	12
20	0.00	0.00	0.00	12
21	0.00	0.00	0.00	3
22	0.00	0.00	0.00	14
23	0.00	0.00	0.00	4
24	0.00	0.00	0.00	12
25	0.12	0.05	0.07	21
26	0.08	0.07	0.07	15
27	0.00	0.00	0.00	3
28	0.00	0.00	0.00	6
29	0.07	0.12	0.09	8
accuracy			0.02	285
macro avg	0.01	0.02	0.01	285
weighted avg	0.02	0.02	0.02	285

HYPERPARAMETER TUNING AND MODEL OPTIMIZATION

Traffic sign images vary widely (shapes, lighting, textures).

A CNN model's accuracy depends heavily on choosing the right settings — too small or too big can lead to:

Underfitting (model too simple)

Overfitting (memorizes training data but fails on new images)

Tuning helps you find the “sweet spot” for best performance.

Accuracy: 0.9262 - loss: 0.3546

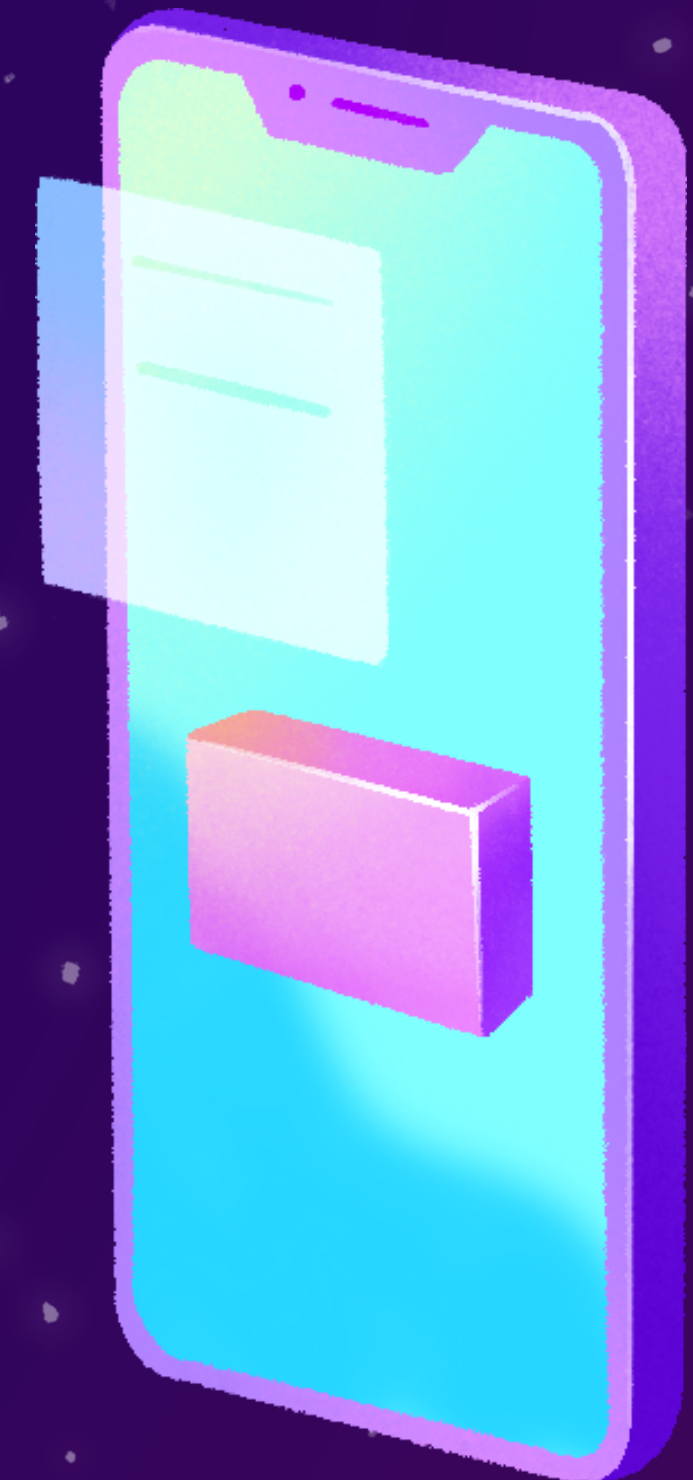
✓ Validation Accuracy: 92.62 %

```
model.save("road_signs_cnn_model.h5")
```

```
print("✓ Model saved as road_signs_cnn_model.h5")
```


STREAMLIT

- A GRAPHICAL USER INTERFACE (GUI) WAS DEVELOPED USING STREAMLIT TO MAKE THE TRAINED CNN ROAD SIGN CLASSIFICATION MODEL INTERACTIVE AND USER-FRIENDLY.
- THE INTERFACE ALLOWS USERS TO UPLOAD ROAD SIGN IMAGES, VISUALIZE THEM, AND INSTANTLY VIEW THE MODEL'S PREDICTIONS ALONG WITH THE ACCURACY SCORE.
- THE STREAMLIT APP INTEGRATES THE TRAINED CNN MODEL (ROAD_SIGNS_CNN_MODEL.H5) AND LABEL FILE (LABELS.CSV).
- WHEN A NEW IMAGE IS UPLOADED, IT IS PREPROCESSED (RESIZED AND NORMALIZED), PASSED INTO THE MODEL, AND THE PREDICTED CLASS NAME AND CONFIDENCE LEVEL ARE DISPLAYED CLEARLY.



STREAMLIT



 Predict Sign

Predicted Sign: Roundabout mandatory

Class ID: 27 | Confidence: 0.36

Predicted Sign: Speed limit (60km/h)

Class ID: 5 | Confidence: 0.50

0:03 / 0:03

INSIGHTS

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- **The CNN-based model effectively learned to recognize and classify various road signs with high accuracy.**
- **Data augmentation improved the model's ability to handle variations such as rotation, brightness, and scale.**
- **The Streamlit interface made the model interactive, allowing real-time testing with new images.**
- **The project demonstrates how deep learning can be applied to enhance intelligent transportation and driver-assistance systems.**

FUTURE SCOPE

- **THE SYSTEM CAN BE EXTENDED TO PERFORM REAL-TIME ROAD SIGN DETECTION USING LIVE VIDEO FEEDS FROM VEHICLE CAMERAS.**
- **INCORPORATING VOICE ALERTS OR DRIVER-ASSISTANCE NOTIFICATIONS CAN ENHANCE SAFETY APPLICATIONS.**
- **EXPANDING THE DATASET TO INCLUDE DIFFERENT COUNTRIES' TRAFFIC SIGNS WILL MAKE THE SYSTEM GLOBALLY APPLICABLE.**

CONCLUSION

The CNN-based Road Sign Recognition system successfully identified and classified traffic signs with high accuracy. The trained CNN model effectively captured essential visual features such as shapes, colors, and patterns of road signs.

Let's Shaping Tomorrow's World

THANK YOU

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